Coffee Orders Sales Dashboard

End-to-End Excel Analytics · Data Cleaning · Pivot Tables · Interactive Dashboard

Portfolio Submission · Data Analytics Project · May 2025

This document presents the design, development, and impact of a business intelligence dashboard built entirely in Microsoft Excel — transforming 1,000 raw coffee orders spanning 3.5 years and 3 international markets into an interactive, filter-driven dashboard.

$45,134	**1,000**	**3**	**4**
Total Revenue	Orders Analysed	Markets	Coffee Types
2019 – 2022	913 unique customers	US · Ireland · UK	Arabica · Excelsa · Liberica · Robusta

01 — Introduction

This project represents a complete data analytics engagement for a fictional coffee retail business operating across the United States, Ireland, and the United Kingdom. The business collected transactional data in Excel but lacked the tools to make sense of it — sales were recorded in one sheet, customers in another, and products in a third, with no connections between them and no way to quickly answer critical business questions.

Dimension	Detail
Data Source	coffeeOrdersData.xlsx — 3 raw worksheets
Time Period	January 2019 – August 2022 (3.5 years)
Order Volume	1,000 transactions, 913 unique customers
Products	48 SKUs — 4 coffee types × 3 roasts × 4 sizes
Markets	United States, Ireland, United Kingdom
Output	coffeeOrdersProject.xlsx — enriched data + dashboard
Tools Used	Microsoft Excel (XLOOKUP, Pivot Tables, Slicers, Charts)

The project follows a standard data analytics workflow: understand the data, clean and join it, aggregate into summaries, and visualise for decision-making — all within a single Excel workbook accessible to any business user.

02 — Business Problem

The coffee retail business faced a common data challenge: **fragmented, disconnected spreadsheet data with no analytical layer on top.** Three core pain points were identified:

PROBLEM 1	**Data Fragmentation**	Orders, customers, and products were stored in separate sheets with no joins or relationships. Answering "Who bought what, where?" required manual VLOOKUP work every time.
PROBLEM 2	**No Revenue Visibility**	The raw orders sheet contained Quantity and Unit Price but no Sales column. Total revenue was unknown without a manual calculation pass across all 1,000 rows.
PROBLEM 3	**No Trend Analysis**	With no pivot tables or charts, spotting seasonal patterns, identifying top markets, or comparing coffee types was impossible without building something from scratch.

Key business question: Which coffee type, market, and customer segment is driving revenue — and how has this changed over time?

03 — Solution Overview

The solution is a single, self-contained Excel workbook that joins all three data sources, calculates revenue, and presents results through an interactive dashboard with slicers and charts — requiring zero coding knowledge to use.

RAW DATA	ENRICHMENT	AGGREGATION	DASHBOARD
orders customers products (3 sheets)	XLOOKUP joins IFS formulas Sales = Qty × Price	Pivot Tables 3 data views (time / country / customer)	Charts + Slicers Timeline filter 1-click insights

Technical approach — Excel-only, no macros, no add-ins:

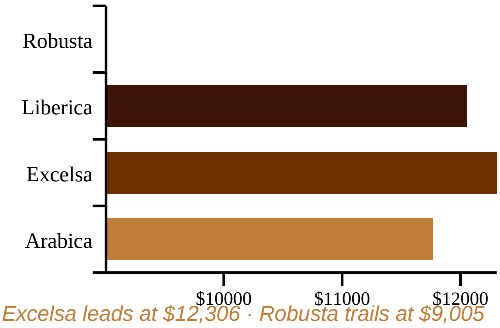
Formula / Feature	Purpose	Example
XLOOKUP	Join customer data to orders	=XLOOKUP(C2,customers!A:A,customers!B:B)
XLOOKUP	Join product data to orders	=XLOOKUP(D2,products!A:A,products!D:D)
Multiplication	Calculate revenue	=E2*L2 (Quantity × Unit Price)
IFS	Expand coffee type codes	=IFS(I2="Rob","Robusta",I2="Exc","Excelsa",...)
Pivot Table	Monthly aggregation	Sum of Sales by Year/Month × Coffee Type
Slicer	Interactive filtering	Roast Type, Package Size, Loyalty Card
Timeline Slicer	Date range filtering	Connected to all 3 Pivot Tables

04 — Key Features & Results

Dashboard Features

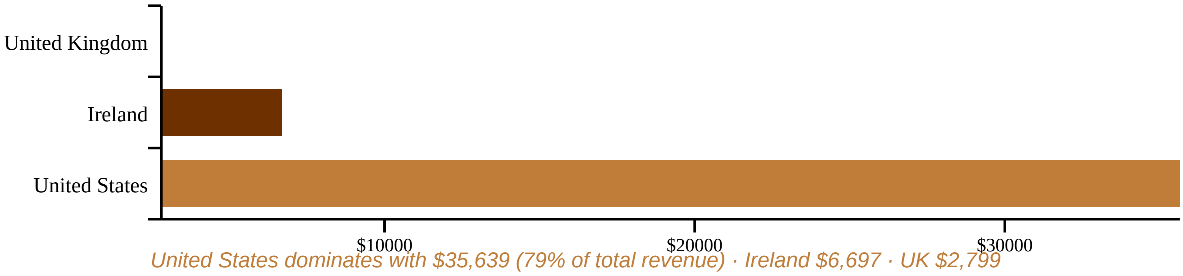
- • Enriched orders sheet with 8 XLOOKUP-joined columns
- • Auto-calculated Sales = Quantity × Unit Price
- • Total Sales over time — line chart by coffee type
- • Revenue by Country — horizontal bar chart
- • Top 5 Customers — ranked bar chart
- • Roast Type slicer — filter all charts at once
- • Package Size slicer — 0.2 / 0.5 / 1.0 / 2.5 kg
- • Loyalty Card slicer — cardholders vs non-holders
- • Timeline slicer — any custom date range

Sales by Coffee Type ($)



Excelsa leads at $12,306 · Robusta trails at $9,005

Revenue by Country



United States dominates with $35,639 (79% of total revenue) · Ireland $6,697 · UK $2,799

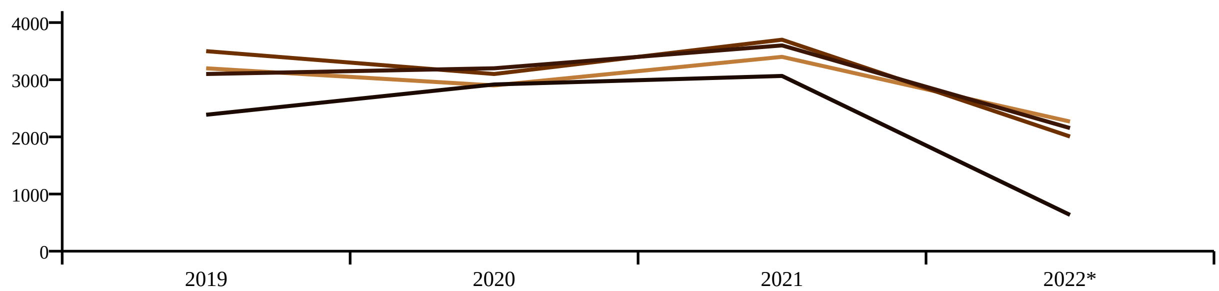
05 — Technical Details

Data Model — Star Schema

Technology Stack

Component	Technology	Version / Notes
Primary Platform	Microsoft Excel	2019+ / Microsoft 365 required
Data Join	XLOOKUP function	Replaces VLOOKUP — handles left lookups
Revenue Calc	Arithmetic formula	=Quantity * Unit Price across 1,000 rows
Text Expansion	IFS function	Decodes abbreviated codes to full names
Aggregation	Pivot Tables	Three separate pivots (time, country, customer)
Visualisation	Pivot Charts	Line chart + 2 horizontal bar charts
Interactivity	Slicers + Timeline	4 interactive filter controls

Yearly Sales Trend



2021 was the strongest year ($13,766). 2022 is partial data through August.

06 — Impact & Results

$45,134	$35,639	$317	$12,306
Total Revenue	US Market Revenue	Top Customer LTV	Top Coffee Revenue
Jan 2019 – Aug 2022	79% of total	Allis Wilmore	Excelsa type

Quantitative Findings

- • Total revenue: **$45,134** across 3.5 years
- • United States: **$35,639 (79%)** of revenue
- • Ireland: **$6,697 (15%)** · UK: **$2,799 (6%)**
- • Excelsa top coffee: **$12,306**
- • Arabica second: **$11,768**
- • Robusta weakest: **$9,005**
- • 2021 best year: **$13,766** revenue
- • 913 unique customers across 1,000 orders
- • 47.9% of customers hold a loyalty card

Business Value Delivered

- • **Eliminated manual joins** — XLOOKUP formulas auto-populate 8 columns instantly for any new order row added
- • **Revenue transparency** — Sales column now auto-calculates; no manual arithmetic needed
- • **Instant filtering** — managers can answer "How did Arabica perform in 2021 for loyalty card holders?" in two clicks
- • **Reusable template** — add new order rows and all charts update automatically via pivot refresh
- • **Shareable** — a single .xlsx file anyone can open and use without special software

Top 5 Customers by Lifetime Spend

Rank	Customer Name	Lifetime Spend
#1	Allis Wilmore	$317.07
#2	Brenn Dundredge	$307.05
#3	Terri Farra	$289.11
#4	Nealson Cuttler	$281.68
#5	Don Flintiff	$278.01

07 — Development Process

The project followed a structured 7-phase development process, mirroring a real-world data analytics engagement from requirements to delivery.

01 **Requirement Gathering — Understanding the Business Problem**
Reviewed both Excel files to catalogue all available fields. Identified the three disconnected tables and the missing relationships. Defined the analytical outputs needed: time-series revenue, geographic breakdown, customer ranking.

02 **Data Audit & Planning**
Mapped which columns in the orders sheet needed to be populated from customers (Customer Name, Email, Country, Loyalty Card) and products (Coffee Type, Roast Type, Size, Unit Price). Planned the XLOOKUP formulas and the derived Sales column.

03 **Data Enrichment — Joining the Tables**
Wrote XLOOKUP formulas for all 8 join columns in the orders sheet. Added the Sales column (=Quantity × Unit Price). Added Coffee Type Name and Roast Type Name columns using IFS to convert abbreviated codes (e.g. "Rob") into readable names (e.g. "Robusta").

04 **Building Pivot Tables**
Created three pivot tables: (1) TotalSales — months as rows, coffee types as columns, sum of sales as values. (2) CountryBarChart — country as rows, sum of sales. (3) Top5Customers — customer name as rows, sum of sales, filtered to top 5.

05 **Building Charts & Visualisations**
Built a line/area chart on the TotalSales pivot to show trends over time. Built horizontal bar charts for country revenue and customer ranking. Applied consistent colour coding across all charts.

06 **Adding Interactivity — Slicers & Timeline**
Inserted a Timeline Slicer connected to all three pivot tables. Added three categorical slicers (Roast Type, Size, Loyalty Card) and connected each to all pivots so filtering one updates all charts simultaneously.

07 **Dashboard Assembly, Testing & Refinement**
Created a dedicated Dashboard sheet. Moved all charts and slicers onto it with a clean layout. Tested all slicer combinations for correct pivot updates. Spot-checked revenue totals against manual calculations. Verified top 5 customer ranking.

08 — Conclusion & Future Potential

This project successfully transforms three disconnected raw spreadsheets into a fully functional, interactive business intelligence dashboard — delivering immediate, actionable insight to coffee retail managers with zero coding requirement.

Goal	Status	Evidence
Join all data sources	Achieved	XLOOKUP across customers & products
Calculate revenue	Achieved	Sales = Qty × Price, all 1,000 rows
Show sales over time	Achieved	Monthly line chart 2019–2022
Compare markets	Achieved	Country bar chart — US dominates at 79%
Identify top customers	Achieved	Ranked top 5 with lifetime spend
Enable self-service filtering	Achieved	4 slicers + timeline on dashboard

Future Enhancement Roadmap

- **Power BI Migration** — Migrate the dashboard to Power BI for web sharing, real-time refresh, and richer interactivity via DAX measures and drill-through pages.
- **Profit Margin Analysis** — The products sheet contains a Profit column. A profitability view overlaying revenue and margin would reveal which coffee types are most valuable, not just highest-selling.
- **Customer Segmentation** — Segment customers by loyalty card status, purchase frequency, and average order value to identify high-value vs at-risk segments.
- **YoY Growth Chart** — Add a Year-over-Year percentage change chart to make growth trends more immediately visible than absolute sales numbers.
- **Automated Data Refresh** — Connect to a live data source (SharePoint list, SQL database) so the dashboard refreshes automatically as new orders are recorded.

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Built with Microsoft Excel · 1,000 Orders · 3 Markets · $45,134 Revenue Analysed